

Basics of artificial intelligence and machine learning

Artificial Intelligence (AI) represents the broad engineering goal of creating computing systems capable of human-like intelligence

. Machine Learning (ML) is the core operational methodology used to achieve modern AI by training systems to learn directly from data.

The Core Hierarchy

Modern intelligent systems are structured as a series of nested sub-disciplines:

- Artificial Intelligence (AI): The comprehensive field covering any technique that enables computers to mimic human cognitive processes, logic, or behavior.
- Machine Learning (ML): A specialized subset of AI that relies on statistical methods to enable algorithms to learn autonomously from experiences and datasets without hard-coded rules.
- Deep Learning (DL): A further specialized branch of ML inspired by the neurological architecture of the human brain, utilizing multi-layered artificial neural networks to decode complex raw features.

2. Primary Machine Learning Paradigms

ML models operate under different learning setups based on data structure and environmental feedback:

Supervised Learning

The system trains on heavily curated, labeled historical data. Every input vector is linked explicitly to a known target output ground truth.

- Key Tasks: Regression (predicting continuous numerical metrics) and Classification (categorizing inputs into discrete classes).

Unsupervised Learning

The network receives completely unlabeled raw inputs. The algorithm must autonomously mine structural relationships, patterns, and groupings within the data.

- Key Tasks: Clustering (grouping similar items) and Dimensionality Reduction (compressing complex feature sets).

Reinforcement Learning

An autonomous software agent operates interactively inside a defined environment. It learns strategic sequences of actions via trial-and-error optimization, aiming to maximize a mathematical reward signal.

Difference between AI, ML and deep learning

Artificial Intelligence (AI): The overarching goal of creating systems or machines that can mimic human intelligence, reasoning, and problem-solving

1. It uses anything from rigid, hard-coded rules and logic trees to advanced data-driven models.
2. Example: Customer service chatbots, chess-playing computers, and automated route planners.

Machine Learning (ML): A core branch of Artificial Intelligence that teaches computers to learn from data without being explicitly programmed for every outcome.

1. Engineers feed data into statistical algorithms (like regression or decision trees), requiring humans to manually label and select specific features.
2. Example: Spam email filters, product recommendation engines, and weather forecasting.

Deep Learning (DL): An advanced subset of Machine Learning that uses multi-layered artificial neural networks modeled after the human brain.

1. It processes massive amounts of raw, unstructured data (like audio or images) and automatically figures out features on its own without human help.
2. Example: Voice assistants, facial recognition, and self-driving cars.

Beginner python programs:

1. **Hello World (Your First Output):** This classic program prints a simple greeting to the screen using the `print()` function.

```
python# Print a simple message to the screen
print("Hello, World!")
```

2. **Add Two Numbers (Using Variables and Input):** This program asks the user for two numbers, adds them together, and shows the result.

```
python# Get input from the user and convert it from text to numbers
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
```

```
# Calculate the sum
total = num1 + num2
```

```
# Display the result
print(f"The sum is: {total}")
```

3. **Even or Odd Checker (Using If-Else Logic)** This program checks if a number entered by the user is even or odd using a conditional statement and the modulo operator (%).

```
python# Get a whole number from the user
number = int(input("Enter an integer: "))
```

```
# Check if the number is divisible by 2
if number % 2 == 0:
    print(f"{number} is Even.")
else:
    print(f"{number} is Odd.")
```